

Reduction types of genus 2 curves

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Models of curves

Throughout, let K be a local field with ring of integers $\mathcal{O}_K$ and residue field k , where $p = \text{char } k > 5$.

Let C/K be a smooth projective curve of positive genus.

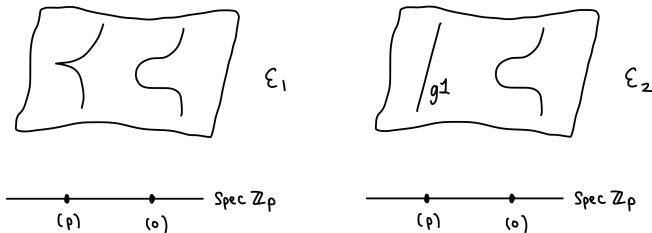
A **model** of C/K is a flat proper scheme $\mathcal{C}/\mathcal{O}_K$ together with a K -isomorphism $\mathcal{C}_\eta \simeq C$. The fibre over the closed point is the **special fibre** $\mathcal{C}_k$.

Example: take $E/\mathbb{Q}_p : y^2 = x^3 + p^6$.

Two projective models of E are given by

$$\mathcal{E}_1 : y^2 = x^3 + p^6$$

$$\mathcal{E}_2 : Y^2 = X^3 + 1.$$



The generic fibres are isomorphic: $x = p^2 X$, $y = p^3 Y$.

Reduction types

A curve C/K has a distinguished model: its **minimal regular model** (MRM). The **reduction type** of C describes the configuration of the special fibre of its MRM.

Informally, the reduction type records the components, their geometric genera, multiplicities and how they meet.

For elliptic curves, these are the **Kodaira types**.

Kodaira symbol	I_0	I_n ($n \geq 1$)	II	III	IV	I_0^*	I_n^* ($n \geq 1$)	IV^*	III^*	II^*
Special fiber $\tilde{C}$ (The numbers indicate multiplicities)										

Why reduction types?

The configuration retains arithmetic information without the full equations of the model.

Write $J = \text{Jac}(C)$. The reduction type gives:

- the **conductor exponent** of J/K .

It measures ramification and enters the completed L -function.

- the **geometric component group** $\Phi(\bar{k})$ of J .

The **Tamagawa number** is $c_K = \#(\Phi(\bar{k})^{\text{Frob}})$.

Local Tamagawa numbers enter the Birch–Swinnerton-Dyer formula.

The case of elliptic curves

Kodaira symbol	I_0	I_n ($n \geq 1$)	II	III	IV	I_0^*	I_n^* ($n \geq 1$)	IV^*	III^*	II^*
Special fiber $\tilde{C}$ (The numbers indicate multiplicities)										
m = number of irred. components	1	n	1	2	3	5	$5 + n$	7	8	9
$E(K)/E_0(K)$ $\cong \tilde{E}(K)/\tilde{E}^0(K)$	(0)	$\frac{\mathbb{Z}}{n\mathbb{Z}}$	(0)	$\frac{\mathbb{Z}}{2\mathbb{Z}}$	$\frac{\mathbb{Z}}{3\mathbb{Z}}$	$\frac{\mathbb{Z}}{2\mathbb{Z}} \times \frac{\mathbb{Z}}{2\mathbb{Z}}$	$\frac{\frac{\mathbb{Z}}{2\mathbb{Z}} \times \frac{\mathbb{Z}}{2\mathbb{Z}}}{\frac{\mathbb{Z}}{4\mathbb{Z}}}$ n even	$\frac{\mathbb{Z}}{3\mathbb{Z}}$	$\frac{\mathbb{Z}}{2\mathbb{Z}}$	(0)
$\tilde{E}^0(K)$	$\tilde{E}(K)$	k^*	k^+	k^+	k^+	k^+	k^+	k^+	k^+	k^+
Entries below this line only valid for $\text{char}(k) = p$ as indicated										
$\text{char}(k) = p$			$p \neq 2, 3$	$p \neq 2$	$p \neq 3$	$p \neq 2$	$p \neq 2$	$p \neq 3$	$p \neq 2$	$p \neq 2, 3$
$v(\mathcal{D}_{E/K})$ (discriminant)	0	n	2	3	4	6	$6 + n$	8	9	10
$f(E/K)$ (conductor)	0	1	2	2	2	2	2	2	2	2
behavior of j	$v(j) \geq 0$	$v(j) = -n$	$\tilde{j} = 0$	$\tilde{j} = 1728$	$\tilde{j} = 0$	$v(j) \geq 0$	$v(j) = -n$	$\tilde{j} = 0$	$\tilde{j} = 1728$	$\tilde{j} = 0$

Table 4.1: A Table of Reduction Types

Genus 2 curves

Every genus 2 curve is **hyperelliptic**, with an affine equation

$$y^2 = f(x), \quad f \in K[x], \quad \deg f = 5 \text{ or } 6, \quad f \text{ separable.}$$

Namikawa–Ueno (1973): **over 100 families** of reduction types.

manuscripta math. 9, 143 - 186 (1973)
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ELLIPTIC TYPE [1]

[Type] [II*-I_n*-m] $\begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & -n \\ \rho & 0 & 1 & 0 \\ 0 & \infty & 0 & 0 & -1 \end{pmatrix}$ m=0 [2] B $\dots$ B $\begin{matrix} 2 \\ 2 \end{matrix} \begin{matrix} 2 \\ 2 \end{matrix} \dots \begin{matrix} 2 \\ 2 \end{matrix}$ n
Mod $\begin{pmatrix} \rho & 0 \\ 0 & \infty \end{pmatrix}$ $\begin{pmatrix} 0 & -1 & 0 & -n \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$ m>0 [39] $\begin{matrix} 3 \\ 4 \\ 5 \end{matrix} \begin{matrix} 4 \\ 3 \\ 4 \end{matrix} \begin{matrix} 2 \\ 4 \end{matrix} \dots \begin{matrix} 2 \\ 2 \end{matrix}$ n
[I₀₋₀-] [I_{6n}-I₀-(6m+8)] $y^2 = (x+t)(x^2+t^{n+2}) \times ((x-1)^3 + t^{6m+5})$
S₂ S₂

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[I₀₋-] [II*-I_n*-α] do. [29] a $\begin{matrix} 2B \\ 3 \\ 4 \\ 5 \end{matrix} \begin{matrix} 4 \\ 3 \\ 4 \end{matrix} \begin{matrix} 2 \\ 4 \end{matrix} \dots \begin{matrix} 2 \\ 2 \end{matrix}$ n
S₂ do. [I_{6n}-I₀-2] $y^2 = t(x^3+t^2)((x-1)^2+t^n)$

[II] [IV-I_n*-m] $\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -n \\ \rho & 0 & -1 & 0 & -1 & 0 \\ 0 & \infty & 0 & 0 & 0 & -1 \end{pmatrix}$ m=0 [2] B $\dots$ B $\begin{matrix} 2 \\ 2 \end{matrix} \begin{matrix} 2 \\ 2 \end{matrix} \dots \begin{matrix} 2 \\ 2 \end{matrix}$ n
 $\begin{pmatrix} z_1 \\ z_3 \end{pmatrix}$ $\begin{pmatrix} \rho & 0 \\ 0 & \infty \end{pmatrix}$ m>0 [39] $\begin{matrix} 3 \\ 4 \\ 5 \end{matrix} \begin{matrix} 4 \\ 3 \\ 4 \end{matrix} \begin{matrix} 2 \\ 4 \end{matrix} \dots \begin{matrix} 2 \\ 2 \end{matrix}$ n
[I_{6n}-I₀-(6m+5)] $y^2 = (x+t)(x^2+t^{n+2})((x-1)^3 + t^{6m+2})$

A genus 2 table

<https://elviralupoian.github.io/genus2reductiontypes/>

Computing the reduction type

For an elliptic curve E/K , the reduction type of E is determined by $v(\Delta_E)$ and $v(j_E)$.

The reduction type of a genus 2 curve can be determined from its *cluster picture* and the valuation of the leading coefficient (Faraggi–Nowell).

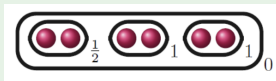
Let $C: y^2 = f(x)$ be a genus 2 curve. The *cluster picture* associated to this model of C displays the p -adic distances between the roots of $f(x)$.

Example

Let $C: y^2 = (x^2 - p)((x - 1)^2 + p^2)((x + 1)^2 - p^2)$ over $\mathbb{Q}_p$. The roots of f are

$$\mathcal{R} = \{\pm\sqrt{p}, 1 \pm ip, -1 \pm p\}.$$

Its cluster picture is



The End

Thank you!